Cindy Shin-Huei Wang* and Cheng Hsiao

Real-Time Monitoring Test for Realized Volatility

Abstract: This paper proposes a monitoring cumulative sum of squares (CUSQ)-type test for structural breaks in *real time* via an autoregressive (AR) approximation framework where data generating process (DGP) is a long memory process. The limiting distribution of the monitoring test follows a Brownian bridge and is free of long memory parameters under the null hypothesis of no break. The test is easy to implement and avoids the issue of spurious breaks found for some retrospective tests for long memory process. Neither does it need to use the bootstrap procedure to find the critical values. Monte Carlo simulations appear to confirm that there exists negligible size distortion and satisfactory power performances in finite samples. The procedure is then applied to monitor the real-time pattern of realized volatilities of dollar–Deutschmark and dollar–Japanese Yen.

Keywords: CUSUM of squares tests, long memory, structural change, real-time, monitoring

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1 Introduction

In this paper, we consider a real-time detector for structural breaks when the underlying process is a long memory process (or an autoregressive fractionally integrated moving average process, denoted as ARFIMA (p, d, q), where d is the fractional differencing parameter). Most tests regarding structural changes are retrospective tests, i.e. given a set of observations, these tests decide if a change has occurred within the time span of the data (e.g. Andrews 1993; Inclan and Tiao 1994). Repeated applications of those retrospective tests could lead to the over-rejection of the true null hypothesis of no break with probability approaching one even there is no break when new observations grow and become available (Robbins 1970). Moreover, some structural break tests for long memory process (e.g. Hildago and Robinson 1996) could have severe size distortion with finite sample, hence may misleadingly point to a structural break while there was none as shown by Kuan and Hsu (1998).

In contrast to the classical one-shot tests to detect a structural break within the data span, Chu, Stinchcombe, and White (1996) suggest an online procedure to monitor structural breaks that has the actual size close to the nominal size. Following their pioneering work, we generalize this procedure to check whether the incoming data are consistent with the estimated model based on the monitoring of cumulative sum of squares (CUSQ) (Brown, Durbin, and Evans 1975) where the underlying process is a long memory process. However, because in finite sample, the maximum likelihood estimator (MLE) cannot discriminate very well between the short- and long-range components in the data when d is close to 1/2 (Crato and Ray 1996), we suggest to approximate a long memory process by an AR model as in Poskitt (2007) and Wang and Hsiao (2013) then construct the CUSUM-type test statistics (denoted as MCUSQAR). We show that the limiting distribution of MCUSQAR follows a Brownian bridge which is free of the long memory parameters. There are several advantages for such a procedure. First, the monitoring CUSQAR test based on the AR spectral estimators is easy to implement, even though the exact order of the ARFIMA (p, d, q) process is *unknown*. Second, the issue of spurious breaks for long memory processes will not arise. Third, there is no need to rely on the bootstrap procedure to deliver a good size performance. Fourth, there is no need to compute different critical values for different long memory parameters and sample size T. Finally, our monitoring test can be used as a test not only for a persistent change in the differencing parameter d but also for a change in the order of integration of a time series, either from stationary short memory processes ($I(0)$ process) to stationary or non-stationary long memory processes or vice versa without assuming that the direction and the location of change are known.

We investigate the finite sample performance of our monitoring tests through Monte Carlo simulations. These results clearly demonstrate that the size and power performance of our MCUSQAR are promising. In other words, there is no indication of the presence of spurious breaks. As an empirical illustration, we monitor the pattern of daily-realized volatility of Deutschemark and Japanese Yen in real time. More specifically, we focus on the question whether, and when, one can gain insight into the evolution of asset volatility patterns and obtain an early indication of upcoming crisis or bubbles using our MCUSQAR test. The empirical findings appear to suggest that our monitoring test is indeed able to detect the real-time pattern of these series.

The outline of this paper is as follows. Section 2 presents the model and hypotheses. Section 3 presents the MCUSQAR test statistics for stationary long memory process. Section 4 provides the evaluation of the size and power of our methodology by Monte Carlo experiments. Section 5 privides an empirical illustration of our monitoring test. Concluding remarks are in Section 5.

2 The model

We consider a real-time structural break test of a linear model of the form

$$y_t = \begin{cases} \mu_o + \eta_{to}, & \text{for} \quad t = 1, \cdots, m, \\ \mu_a + \eta_{ta}, & \text{for} \quad t \geq m+1. \end{cases}, \qquad [1]$$

where μ is a regression coefficient and η is a long memory process, $I(d)$ or an ARFIMA (p, d, q) satisfying:

Assumption 1. *η_t is generated as:*

$$\phi(L)(1-L)^d \eta_t = \theta(L) e_t, \qquad [2]$$

where (i) $d \in (-0.5, 0.5)$; (ii) $(1-L)^d$ is the fractional differencing operator defined as $(1-L)^d = \sum_{j=0}^{\infty} \beta_j L^j$; (iii) the AR and MA-polynomials $\phi(L) = \sum_{j=0}^{p} \phi(j) L^j$ and $\theta(L) = \sum_{j=0}^{q} \theta_j L^j$ are assumed to have all roots outside the unit circle; (iv) $\phi(L)$ and $\theta(L)$ have no common zeroes; (v) e_t is an independently and identically distributed process, with $E(e_t) = 0$, $E(e_t^2) = \sigma^2$, and $E\left(e_t^4\right) < \infty$.

The null and alternative hypotheses of interest are

$H_0 : \mu_{ta} = \mu_{to}$ and the distribution of η_{ta} is the same as that of η_{jo} for $t \geq m+1$ and $j = 1, 2, \ldots, m$.

$H_1 : \mu_{ta} \neq \mu_{to}$ and/or the distribution of η_{ta} differs from that of η_{jo} for $t \geq m+1$ and $j = 1, 2, \ldots, m$.

In order words, we could consider three types of breaks: (i) a shift in the regression coefficient μ, (ii) a shift in the distributions of the error term η_t, such as a change of their variance (equal to a shift in long memory parameters d and/or in AR and/or MA coefficients), and (iii) some combinations of shifts in μ and/or η_t.

3 MCUSQAR test

Following the pioneering work of Chu Stinchcombe, and white (1996), we generalize the CUSQ test as an online procedure to monitor structural breaks for a long memory process. As described in Chu, Stinchcombe, and white (1996), given an initial fixed training sample of size m, the monitoring scheme needs a stopping device, determined by a detecting statistics (detector) T_n, and a boundary function $g(n/m)$, to monitor whether the breaks occur after the data being increased to $n(n \geq m)$. More precisely, under the hypothesis of no change, T_n may cross a boundary function $g(n/m)$, for some $n \geq m$, with certain probability, say 0.05 or 0.10. On the other hand, if a break indeed occurs, we expect T_n to cross $g(n/m)$ with a large probability. Operationally, the null hypothesis is rejected when $T_n \geq g(n/m)$ for some $n > m$. Otherwise, the monitoring process keeps on running until a break is observed.

Following Brown, Durbin, and Evans (1975); Inclan and Tiao (1994); Chu Stinchcombe, and white (1996), we propose to use the centered version of the CUSQ of the (uncorrelated) innovation e_t to construct the detectors, T_n, for an $I(d)$ process satisfying Assumption 1. Brockwell and Davis (1991) have shown that for η_t under Assumption 1 can be represented by an infinite-order AR process,

$$\eta_t = \sum_{j=1}^{\infty} \beta_j \eta_{t-j} + e_t, \tag{3}$$

where

$$\beta_j = \frac{\Gamma(j-d)}{\Gamma(j+1)\Gamma(d)}, \ d \in (-0.5, 0.5)$$

and e_t is a zero-mean white-noise process with constant variance σ^2. Poskitt (2007) further shows that η_t can be approximated by an ever increasing order AR model, AR(k), as time series observation T increases,

$$\eta_t = e_{tk^*} + \sum_{j=1}^{k} \beta_{jk} \eta_{t-j}, \tag{4}$$

where e_{tk} is the prediction error and β_{jk}, $j = 1, 2, \cdots, k$ are the coefficients of the minimum mean squared predictor of y_t based only on the past observations $\eta_{t-1,t-2}, \cdots, \eta_{t-k}$. Let $\beta^*(k) = (\beta_{1k}, \cdots, \beta_{kk})'$, $\widehat{\beta}^*(k)$ be the ordinary least square (OLS) estimates of $\beta(k)$ and $\widehat{e}_{tk^*}$ be the residuals of the least squares regression model of [4]. Generalizing the results of Poskitt (2007), Wang and Hsiao (2013)[1] proposed the following Lemma.

Lemma 1. *Consider an AR(k) approximation for the data generating process (DGP) satisfying Assumption 1, where k increases with the initial training sample size m and $k/m \to 0$. As $m \to \infty$, uniformly*

1. $\widehat{\sigma}^2_{t,k} = \frac{1}{m}\sum_{t=k+1}^{m} \widehat{e}^2_{tk^*} \xrightarrow{p} \sigma^2$, when $d \in (-0.5, 0.5)$.

Therefore, we propose a detector, T_n, which is based on the CUSUM of squares statistic originally proposed by Brown, Durbin, and Evans (1975) and Inclan and Tiao (1994) to test for a change in the variance of an i.i.d. sequence. Following the analysis of Inclan and Tiao (1994), we let $C_l = \sum_{t=1}^{l} e_t^2$ be the CUSQ of a series of uncorrelated random variable e_t with mean 0 and variance σ_e^2, $t = 1, 2, \ldots$, and denote the centered (for $n \geq m$) CUSQ as

$$D_n = \frac{C_n}{C_m} - \frac{n-2}{m-2}.$$

Let

$$\begin{aligned} S_n = 2^{-1/2} m D_n &= 2^{-1/2} \sum_{t=1}^{n} \left(\frac{e_t^2}{\widehat{\sigma}_e^2} - 1 \right) - 2^{1/2}(n-m)/(m-2) \\ &= 2^{-1/2} \sum_{t=m+1}^{n} \left(\frac{e_t^2}{\widehat{\sigma}_e^2} - \frac{m}{m-2} \right) \end{aligned}$$

where $\widehat{\sigma}_e^2 = \frac{1}{m}\sum_{t=1}^{m} e_t^2$ is the estimator of σ^2 based on historical dataset, $e_1, e_2, \cdots, e_m$.

We further let

$$\widehat{S}_n = 2^{-1/2} m \widehat{D}_n \text{ with } \widehat{D}_n = \frac{\widehat{C}_n}{\widehat{C}_m} - \frac{n-2}{m-2} \text{ and } \widehat{C}_l = \sum_{t=k+1}^{l} \widehat{e}^2_{tk}.$$

Thus, under some conditions on the growth rate of the lag length k, we have [2,3] the following lemma.

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1 The methodology they used for proving Lemma 1 here is different from that of Poskitt (2007); thus, the growth rate condition of k used by them is more relaxing.

2 In Appendix, we show that when $0 < d < 0.5$ and $k = O(m^{0.8})$, the Lemma 2 holds as well. But for simplicity, we relax the condition of k for both cases in the Lemma 2.

3 Our simulation results appear to indicate that the same results hold for $-0.5 < d < -0.25$. However, the proof of this more general result is nontrivial and it will be left for future study.

Lemma 2. *Consider an AR(k) approximation for the DGP satisfying Assumption 1, where k increases with the historical sample size m and is set proportional to the sample size, i.e. k = O(m) or k = τm, 0 < τ < 1. As n, m → ∞ and* $\frac{n}{m} = b > 1$ *is finite,*

1. $m^{-1/2}\widehat{S}_n = 2^{-1/2}m^{-1/2}\sum_{t=m+1}^{n}\left(\frac{\widehat{e}_{tk}^2}{\widehat{\sigma}_k^2} - \frac{m}{m-2}\right) = m^{-1/2}S_n + o_p(1),$ when $d \in (-0.25, 0) \cup (0, 0.5)$ and $\widehat{\sigma}_k^2 = \frac{1}{m}\sum_{t=k+1}^{m}\widehat{e}_{tk}^2$.

The proof is given in the Appendix.

By the Proposition 6.1.2 and item (iii) of the Definition 6.1.4 of Brockwell and Davis (1991), we further note that

$$m^{-1/2}\widehat{S}_n \xrightarrow{p} m^{-1/2}S_n.$$

Lemma 1 and 2 imply that the residuals of an AR(k) approximation of a long memory processes are i.i.d asymptotically. Hence, according to Theorem 1 of Inclan and Tiao (1994), we know that under the null hypothesis H_0,

$$T_n = m^{-1/2}\widehat{S}_{[m\tau]} \Longrightarrow W^0(\tau),$$

where "$\Longrightarrow$" denotes weak convergence, $[m\tau] = n$, $\tau \geq 1$ and $W^0(\tau) = W(\tau) - \tau W(1)$, with $W(\tau)$ being a standard Wiener Process, and $W^0(\tau)$ is a Brownian bridge.

By the same reasoning as Chu, Stinchcombe, and White (1996), the crossing probability of $W^0(\tau)$, $\tau \geq 1$ can be investigated in terms of the Wiener process. We choose $g(\tau) = [\tau(a^2 + ln(\tau))]^{1/2}$ to be our boundary function and put $\tau = \frac{\lambda}{\lambda-1}$, then as shown in (13) in Chu, Stinchcombe, and White (1996),

$$\begin{aligned} P\{|W^0(\tau)| = (\tau-1)^{-1}|W(\tau)| \geq (\tau-1)^{-1}[\tau(a^2+ln(\tau))]^{1/2} \\ \text{for some } \tau \geq 1\} \\ = 2[1-\Phi(a)+a\phi(a)], \end{aligned} \quad [5]$$

where Φ and ϕ are the cdf and pdf, respectively, of a standard normal random variable. The crossing probability can be easily computed via the right-hand side of [5]. When $a^2 = 7.78$, the crossing probability is 5%. Discretizing λ as n/m yields the boundary

$$g(\frac{n}{m}) = (\frac{n-m}{m})[(\frac{n}{n-m})[a^2 + ln(\frac{n}{n-m})]]^{1/2}.$$

It follows that

$$lim_{m\to\infty} p\{|\widehat{S}_{[m\tau]}| \geq m^{1/2}(\frac{n-m}{m})[(\frac{n}{n-m})[a^2 + ln(\frac{n}{n-m})]]^{1/2},$$

$$\text{for some } n \geq m\},$$

$$\cong 2[1 - \Phi(a) + a\phi(a)].$$

The asymptotic properties of the detecting test statistic T_n are summarized in Theorem 1.

THEOREM 1. *Consider $y_t = \mu_t + \eta_t$ where η_t satisfies Assumption 1. When the conditions of Lemma 2 hold, the following functional central limit (FCLT) holds for MCUSQAR test statistic, as n, m → ∞ and $\frac{n}{m} = b > 1$ is finite, under the null hypothesis of no structural change, T_n:*

$$T_n \Rightarrow W^0(t) = W(t) - tW(1),$$

where W and W^0 are the (one-dimensional) Brownian motion and Brownian bridge, respectively, such that

$$lim_{m\to\infty} P\{|T_n| \geq g(t), for\ some \quad t \geq 1\} = 2[1 - \Phi(a) + a\phi(a)]. \qquad [6]$$

Theorem 1 implies that our MCUSQAR test is easy to implement and its asymptotic property does not depend on distribution of the error term η_t, including the long memory parameter d and the coefficients of MA and AR terms. Additionally, we note that the test T_n can be applied to models estimated not only by OLS but also by instrumental variables and generalized method of moments.

The MCUSQAR test can also be applied to nonstationary long memory processes. Suppose that the DGP is of the form

$$y_t = y_{t-1} + \eta_t, \qquad [7]$$

where η_t is a stationary long memory process (or an $I(0)$ process) and y_t is a nonstationary long memory process (or an $I(1 + d)$ process).

Taking the first difference of y_t, we will have

$$\triangle y_t = \eta_t, \qquad [8]$$

where $\Delta = (1 - L)$ and L is the lag operator. Using the same argument as the stationary long memory case, we can approximate Δy_t by an AR(k) model

$$\eta_t = \sum_{j=1}^{k} \widehat{\beta}_{jk}\eta_{t-j} + \widehat{e}_{tk}, \qquad [9]$$

where the growth rate of the order, k, satisfies that given in Lemma 1. Accordingly,

Theorem 2. *Let y_t be the processes given by [7] and η_t satisfies Assumption 1. When the conditions of Lemma 2 hold and $n \to \infty$, the following FCLT holds for MCUSQAR test statistics under the null hypothesis of no structural change:*

$$T_n \Rightarrow W^0(t) = W(t) - tW(1),$$

where W and W^0 are the (one-dimensional) Brownian motion and Brownian bridge, respectively, such that

$$lim_{m\to\infty} P\{|T_n| \geq g(t), \text{for some} \quad t \geq 1\} = 2[1 - \Phi(a) + a\phi(a)]. \qquad [10]$$

Theorem 2 says that the limiting distribution of MCUSQAR test T_n also follows the Brownian bridge when the DGP is a nonstationary long memory process. In other words, the MCUSQAR test can be also applied as a test for a change in the order of integration of a time series either from $I(d)$ to $I(1 + d)$ or from $I(1 + d)$ to $I(d)$, $d \in (-0.25,0.5)$.

4 Monte Carlo simulation

This section examines the finite sample behavior of the proposed monitoring procedure through Monte Carlo simulation experiments. We simulate two ARFIMA (p, d, q) processes of the form:

$$\begin{aligned} &\text{DGP (a)}. (1 - L)^d \eta_t = e_t, \\ &\text{DGP (b)}. (1 - \phi_1 L)(1 - L)^d \eta_t = (1 + \theta_1)e_t \\ &\qquad\qquad (\phi_1 = -0.7, \theta_1 = 0.5), \end{aligned}$$

with differencing parameter d = 0.1, 0.2, 0.3, 0.4, 0.45, 0.49, 0.499.

We follow the experiment designs of Chu, Stinchcombe, and White (1996). More specifically, we consider four training sample size m (m = 50, 100, 200, and 300) and three different monitoring horizons q (q = 2m, 3m, and 4m). According to our Theorems and following the simulation suggestion of Sun (2004), we consider the values of τ = 0.02, 0.04, 0.06, 0.08, and 0.1 to choose the lag order $k = \tau m$ of the $AR(k)$ approximation for the long memory process η_t. Because the results are similar, we only focus our discussion on the case of $k = \tau m$, τ = 0.04 and with two types of breaks in our simulation: (i) breaks in μ and (ii) breaks in μ or/and differencing parameters d's only. Results on the case of $k = \tau m$, τ = 0.02, 0.06, 0.08, and 0.1 and on the case of the breaks in AR and MA coefficients are not reported here but are available on request. All the simulations are based on 2,500 replications.

Table 1: Empirical size of T_n, when the DGP of η_t is either (a) or (b).

DGP	*m*	*q*	0.1	0.2	0.3	0.4	0.45	0.49	0.499
(*a*)	50	2*m*	4.9	5.3	5.7	6.0	6.3	6.5	6.7
		3*m*	5.3	5.8	6.0	6.2	6.3	6.7	6.9
		4*m*	5.7	6.1	6.2	6.4	6.6	7.0	7.1
	100	2*m*	4.5	4.7	4.9	4.8	4.0	5.3	5.4
		3*m*	4.6	4.7	5.2	5.2	5.5	5.9	6.1
		4*m*	4.9	4.9	5.3	5.4	5.7	6.1	6.1
	200	2*m*	4.0	4.2	4.2	4.5	4.5	4.6	4.9
		3*m*	4.1	4.3	4.2	4.5	4.7	4.8	4.8
		4*m*	4.2	4.4	4.5	4.7	4.6	4.9	5.2
	300	2*m*	3.0	3.2	3.4	3.6	4.1	4.4	4.5
		3*m*	3.2	3.3	3.6	3.8	4.2	4.8	4.7
		4*m*	3.5	3.7	3.9	4.1	4.6	4.7	4.9
(*b*)	50	2*m*	5.4	5.7	6.0	6.2	6.4	7.0	7.1
		3*m*	5.6	6.0	6.3	6.5	6.9	7.3	7.5
		4*m*	6.0	6.2	6.5	6.6	7.0	7.4	7.7
	100	2*m*	4.6	4.7	5.1	4.9	5.1	5.7	5.7
		3*m*	4.8	4.8	5.2	5.2	5.5	5.9	6.0
		4*m*	5.0	5.1	5.3	5.4	5.7	6.1	6.3
	200	2*m*	4.2	4.4	4.6	4.5	4.5	4.6	4.9
		3*m*	4.2	4.5	4.6	4.6	4.7	4.8	4.8
		4*m*	4.5	4.6	4.7	4.9	4.6	4.9	5.2
	300	2*m*	3.7	3.2	3.4	3.6	4.1	4.4	4.5
		3*m*	3.6	3.3	3.7	3.9	4.2	4.8	4.7
		4*m*	3.9	3.7	4.2	4.1	4.6	4.7	4.9

Tables 1 displays the frequencies of rejection of the null hypothesis of no structural change for some n lying between m and q for the monitoring test T_n and DGPs (a) and (b) with nominal size of 5%, which corresponds to choosing $a^2 = 7.78$ in [6]. They give no hint of improper size distortions.

To investigate the finite sample power performance of our monitoring test procedure, we create two types of out-of-sample structural breaks: in the regression coefficient μ of [1] and in the structure of the error term process η_t, including a break in the long memory parameter d. First, we create an artificial break at time $t = m \times 1.1$, at which the regression coefficient permanently shifts from 2 to 2.8. In Table 2, we report the power values for this break configuration. Indeed, our monitoring procedure managed to signal the structural change. Second, we consider 10 types of break in the DGP of the error process η_t, as listed in Table 3. We also examine a structural shift from a standard normal process to a stationary long memory process as experiments 1 and 2 in Table 3. We denote the DGP before the break and after the break as DGP(I) and DGP(II),

Table 2: Power of T_n, when the DGP of η_t is either (a) or (b).

DGP	***m***	**q**	**0.1**	**0.2**	**0.3**	**0.4**	**0.45**	**0.49**	**0.499**
(*a*)	50	2*m*	19.5	21.0	20.4	24.7	26.4	41.1	55.2
		3*m*	22.6	24.9	22.9	28.9	33.8	50.2	58.2
		4*m*	26.4	37.1	29.2	37.5	41.2	54.8	60.1
	100	2*m*	34.3	44.7	53.0	62.2	68.5	69.9	70.1
		3*m*	50.2	57.5	67.7	74.9	77.2	83.4	92.6
		4*m*	66.7	70.9	72.3	81.2	85.7	90.1	93.0
	200	2*m*	54.1	64.2	70.1	77.9	85.2	89.1	92.2
		3*m*	62.3	70.1	77.2	84.5	90.6	94.8	100
		4*m*	72.6	78.2	81.3	91.7	98.2	100	100
	300	2*m*	63.7	70.2	79.1	88.0	94.1	92.4	96.5
		3*m*	77.5	77.3	87.2	93.2	97.9	98.8	99.5
		4*m*	80.0	82.9	90.9	98.8	100	100	100
(*b*)	50	2*m*	21.2	22.0	23.4	28.7	30.4	43.1	58.2
		3*m*	27.6	28.9	28.9	30.9	33.8	52.2	62.4
		4*m*	29.1	30.1	33.2	33.5	41.2	57.9	63.1
	100	2*m*	37.1	48.2	54.1	63.7	67.2	70.9	74.5
		3*m*	56.3	59.3	70.0	77.9	80.1	86.0	93.1
		4*m*	67.2	72.9	74.3	82.2	88.8	95.1	94.7
	200	2*m*	56.0	65.2	72.1	77.9	66.2	93.2	96.0
		3*m*	66.1	71.1	80.2	89.6	91.6	95.1	100
		4*m*	74.0	80.2	84.3	94.7	99.9	100	100
	300	2*m*	65.1	72.1	81.5	89.6	95.2	97.4	99.5
		3*m*	78.2	79.6	88.1	95.1	99.9	100	100
		4*m*	83.2	86.1	92.1	98.8	100	100	100

Table 3: Breakpoint specifications by experiments (EX).

EX	***DGP(I)***	***DGP(II)***	d_1	d_2	μ_1	μ_2
1	*ND*	(*a*)	0.0	0.1	0.0	0.0
2	*ND*	(*b*)	0.0	0.1	0.0	0.0
3	(*a*)	(*a*)	0.3	0.45	0.0	0.0
4	(*a*)	(*a*)	0.3	0.45	0.3	0.9
5	(*b*)	(*b*)	0.3	0.45	0.3	0.9
6	(*b*)	(*b*)	0.3	0.45	0.0	0.0
7	(*a*)	(*b*)	0.3	0.45	0.0	0.0
8	(*a*)	(*a*)	1.3	1.45	0.0	0.0
9	(*b*)	(*b*)	1.3	1.45	0.0	0.0
10	(*a*)	(*b*)	1.3	1.45	0.0	0.0

Note: *ND* ~ $N(0, 1)$.

respectively. All corresponding size and power values are reported in Table 4 for T_n. These results demonstrate that the size distortions are small and the power performances are promising. More specifically, the corresponding size and power results of specific experiments 1 and 2 imply that our monitoring test can also be applied to a test of a short memory process ($I(0)$ process) against a long memory process ($I(d)$ process).

5 Application to realized volatility

In this section, we illustrate the application of the monitoring procedure T_n to realized volatility of the foreign exchange rate. Following Andersen et al. (2003), we estimate realized volatility by a consistent nonparametric estimation method. This error-free estimation of volatility would allow us to treat volatility as an observable, rather than a latent variable as with a GARCH(1,1) model for example. Moreover, Anderson et al. (2003) find logarithmic realized volatility could be modeled and accurately forecasted using simple parametric fractional integrated ARFIMA models as [2].

We use the same data as Andersen et al. (2003) and Choi, Yu, and Zivot (2010), which are spot exchange rates for the US dollar, the Deutschemark and the Japanese Yen from 1 December 1986 through 30 June 1999. This data set consists of 3,045 days of intra-day 30-minute log returns on the DM/USD and Yen/USD exchange rates. The statistics and the patterns of the two log realized volatilities were analyzed in detail in Table 1 and Figure 4 of Choi, Yu, and Zivot (2010). We denote the two log realized volatility series as Y_{DM} and Y_{JN}, respectively. We use a model with constant mean (i.e. no trend is included) to estimate the OLS residuals. These residuals are then fitted by an AR model. The residuals of this AR model are the basic inputs of our test statistic. We let the training sample size $m = 100$. Because the estimated differencing parameters of both series are positive (see Table 1 of Choi, Yu, and Zivot 2010), the lag orders k of the AR approximation are based on $k = \tau m$, $\tau = 0.04$. The monitoring starts with the 101th observation and is interrupted if a break is found at some later date. Then it starts again with a training sample of 100 initial observations after the break date until a new break is found. We stop when there are less than m observations left at the end of the sample, because our training sample to estimate the model is set to m.

In Table 5, we report the break dates detected by our monitoring procedure. The break dates detected for Y_{DM} and Y_{JN} are very similar. We detect seven and eight breaks for Y_{DM} and Y_{JN}, respectively. More specifically, compared to the

Table 4: The size and power of T_n test for experiments designed in table 3.

	m	q	*EX* 1	2	3	4	5	6	7	8	9	10
Size	50	2*m*	5.4	5.6	5.8	5.9	6.3	6.0	6.6	6.2	6.7	6.9
		3*m*	5.6	5.7	6.0	5.9	6.3	6.2	7.0	6.3	6.7	7.2
		4*m*	5.9	6.1	6.1	6.2	6.7	6.4	7.2	6.6	7.0	7.5
	100	2*m*	5.1	5.3	5.5	5.6	6.0	5.6	6.3	6.0	6.4	6.9
		3*m*	5.2	5.5	5.9	5.7	6.2	5.8	6.7	6.3	6.9	7.0
		4*m*	5.2	5.7	5.9	5.9	6.4	6.0	7.0	6.5	6.9	7.2
	200	2*m*	4.7	5.3	5.3	5.2	5.6	5.3	5.6	5.5	5.9	6.1
		3*m*	5.2	5.4	5.5	5.5	5.9	5.6	5.8	5.6	6.1	6.2
		4*m*	5.2	5.7	5.6	5.7	5.8	5.8	5.9	6.0	5.9	6.3
	300	2*m*	3.9	4.2	4.0	4.2	4.7	4.5	5.1	5.7	5.3	5.8
		3*m*	4.0	4.6	4.5	4.6	5.0	4.8	5.4	5.3	5.7	6.0
		4*m*	4.0	4.5	4.6	4.7	5.0	4.8	5.6	4.9	5.8	6.1
Power	50	2*m*	18.2	20.0	26.4	22.1	36.4	44.1	51.2	18.5	51.1	58.8
		3*m*	19.6	20.9	30.0	29.9	38.8	47.2	53.2	29.4	58.2	60.1
		4*m*	22.1	27.1	36.9	39.4	42.2	52.7	63.1	32.5	60.1	67.2
	100	2*m*	21.2	28.4	29.2	30.2	40.1	50.1	56.7	31.9	60.1	64.6
		3*m*	29.3	30.9	34.7	34.1	42.2	59.4	60.6	32.2	66.5	69.4
		4*m*	31.7	36.2	38.3	38.2	44.3	67.9	72.2	39.5	69.9	76.1
	200	2*m*	24.2	30.1	40.1	47.9	52.7	59.1	62.2	38.2	66.7	70.2
		3*m*	30.2	38.1	44.2	54.5	56.6	64.8	69.7	50.7	69.5	76.5
		4*m*	32.6	40.2	52.3	61.7	61.2	70.2	79.1	59.8	77.5	78.4
	300	2*m*	33.6	45.2	55.1	60.2	72.7	77.4	89.1	69.1	80.1	89.0
		3*m*	37.5	57.1	69.2	71.2	80.0	85.4	90.2	72.6	84.2	96.5
		4*m*	40.2	67.9	75.9	78.3	84.2	89.0	100.0	82.1	89.0	100

Table 5: Multiple structural changes dates detection.

Series	T_1	T_2	T_3	T_4	T_5
DM/USD	1987/9/27	1989/3/28	1991/3/15	1992/10/8	1994/4/9
YEN/USD	1987/9/27	1989/5/10	1991/3/27	1992/5/25	1994/4/9
Series	T_6	T_7	T_8	T_9	T_{10}
DM/USD	1995/8/7	1997/7/27			
YEN/USD	1995/5/18	1997/5/15	1998/12/15		

finding of Choi et al. (2010), we get more breaks for both series. For example, they only found five breaks for both series. Two common break dates detected by our monitoring procedure for both series are 27 September 1987 (close to stock market crash of October 1987) and 9 April 1994 (before the Mexican Peso crisis of December 1994). Two particular dates found are 15 May 1997 for Japanese Yen (before the Asian financial crisis of July 1997) and 27 July 1997 for Deutschemark (close to the Asian financial crisis). In other words, our monitoring procedure appears to be able to capture the signal of the financial crisis. The one more break date monitored for Y_{JN} is 15 December 1998, a date prior to the break out of the Latin America financial crisis in January 1999). These break dates seem to make sense and to match the events. In particular, the evolving events appear to corroborate the findings of our monitoring test.

6 Concluding remarks

This paper suggests a monitoring CUSQ-type test in *real-time* to check whether the incoming data are consistent with the existing estimated model via an AR approximation when DGPs are long memory processes and when the exact form of the long memory processes are unknown. We have shown that the limiting distribution of our monitoring test follows a Brownian bridge and the test statistic does not depend on the exact nature of long memory processes. The procedure can be implemented without the need to conduct the bootstrap procedure. Simulation results show that the proposed monitoring procedure is able to detect the structural change without size distortion and has promising power performance in finite sample. We have also applied our methodology to monitor the real-time pattern of realized volatilities of two daily nominal dollar exchange rates, Deutschemark and Japanese Yen. The empirical findings appear to demonstrate that our monitoring tests indeed can capture the real-time pattern of the exchange rate realized volatilities before and after the financial crisis.

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Appendix

A.1 Proof of Lemma 2

To prove the Theorem 1, some of the arguments that follow are similar to those in Ng and Perron (2005). We first define

$$\beta(k) = (\beta_1, \beta_2, \ldots, \beta_k), \quad \beta(k+1, \infty) = (\beta_{k+1}, \beta_{k+2}, \ldots,),$$

$$\zeta_k = \begin{bmatrix} \eta_{(m+1)-1} & \eta_{(m+1)-2} & \cdots & \eta_{(m+1)-k} \\ \eta_{(m+2)-1} & \eta_{(m+2)-2} & \cdots & \eta_{(m+2)-k} \\ \vdots & \vdots & \vdots & \vdots \\ \eta_{n-1} & \eta_{n-2} & \cdots & \eta_{n-k} \end{bmatrix}_{(n-m)\times k}$$

$$= (\widetilde{\eta}_1, \widetilde{\eta}_2, \ldots, \widetilde{\eta}_k),$$

$$\zeta_{k^+} = \begin{bmatrix} \eta_{(m+1)-(k+1)} & \eta_{(m+1)-(k+2)} & \cdots & \cdots \\ \eta_{(m+2)-(k+1)} & \eta_{(m+2)-(k+2)} & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \eta_{n-(k+1)} & \eta_{n-(k+2)} & \cdots & \cdots \end{bmatrix}$$

$$= (\widetilde{\eta}_{k+1}, \widetilde{\eta}_{k+2}, \ldots,)$$

$$\varepsilon_{n-m} = (e_{m+1}, e_{m+2}, \cdots, e_n)',$$

$$\widetilde{\eta} = (\eta_{m+1}, \eta_{m+2}, \ldots, \eta_n)',$$

where both ε_{n-m}, $\widetilde{\eta}$ and all $\widetilde{\eta}_1, \widetilde{\eta}_2, \cdots, \widetilde{\eta}_{k+1}, \cdots$ are $(n-m)\times 1$ vectors, respectively. Since n is the monitoring horizon, we set $n = bm$ with $b > 1$ and b is a real number and a positive constant.

Suppose the true model is

$$\widetilde{\eta} = \zeta_k \beta(k)' + \zeta_{k^+}\beta(k+1, \infty)' + \varepsilon_{n-m}.$$

We cannot estimate an infinite number of parameters from a finite sample of m observations. We consider the following kth-order AR model approximation:

$$\widetilde{\eta} = \zeta_k \beta(k)' + \varepsilon_k, \tag{A.1}$$

where $e_{t,k}$, $t = m + 1, m + 2, \cdots, n$ is the residual derived from fitting η_t by an AR (k) approximation, and $\varepsilon_k = (e_{m+1,k}, e_{m+2,k}, \cdots, e_{n,k})'$. The adequacy of the approximate model for the true model depends on the choice of k.

Let $M_k = I - \zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'$ be an $(n-m)\times(n-m)$ symmetric, idempotent matrix with the property that $M_k\zeta_k = 0$, then the OLS estimate of the residual of [A.1] ε_k can be written as

$$\begin{aligned}\widehat{\varepsilon}_k &= M_k\left(\varepsilon_{n-m} + \sum_{h=1}^{k}\tilde{\eta}_h\beta_h + \sum_{h=k+1}^{\infty}\tilde{\eta}_h\beta_h\right)\\ &= M_k\zeta_{k+}\beta(k+1,\infty)' + M_k\varepsilon_{n-m}.\end{aligned}$$

We wish to show that when $-0.25 < d < 0.5$,

$$\begin{aligned}\widehat{Q} &= m^{-1/2}\widehat{\varepsilon}_k'\widehat{\varepsilon}_k\\ &= m^{-1/2}(\beta(k+1,\infty)\zeta_{k+}'M_k\zeta_{k+}\beta(k+1,\infty))'\\ &+ \beta(k+1,\infty)\zeta_{k+}'M_k\varepsilon_{n-m} + \varepsilon_{n-m}'M_k\zeta_{k+}\beta(k+1,\infty)'\\ &+ \varepsilon_{n-m}'M_k\varepsilon_{n-m})\\ &= m^{-1/2}\beta(k+1,\infty)\zeta_{k+}'M_k\zeta_{k+}\beta(k+1,\infty)' \qquad [A.2]\\ &+ m^{-1/2}\beta(k+1,\infty)\zeta_{k+}'M_k\varepsilon_{n-m} + m^{-1/2}\varepsilon_{n-m}'M_k\zeta_{k+}\beta(k+1,\infty)'\\ &+ m^{-1/2}\varepsilon_{n-m}'\varepsilon_{n-m} - m^{-1/2}\varepsilon_{n-m}'\left(\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\right)\varepsilon_{n-m}\\ &= (I) + (II) + (III) + (IV) + (V)\\ &= m^{-1/2}\varepsilon_{n-m}'\varepsilon_{n-m} + o_p(1).\end{aligned}$$

That is, we need to find out a condition on growth rate of lag order k to ensure that the first term (I), the second term (II), the third term (III), and the fifth term (V) of [A.2] converge to zero when $-0.25 < d < 0.5$.

We then investigate the property of [A.2] for the case of $0 < d < 0.5$. The first term (I) on the right-hand side of [A.2] can be written as

$$(I) = m^{-1/2}(\beta(k+1,\infty)\zeta_{k+}'M_k\zeta_{k+}\beta(k+1,\infty)')$$

$$
\begin{aligned}
&= m^{-1/2}\{\beta(k+1,\infty)\zeta'_{k+}I\zeta_{k+}\beta(k+1,\infty)'\}\\
&- m^{-1/2}\{\beta(k+1,\infty)\zeta'_{k+}\zeta_k(\zeta'_k\zeta_k)^{-1}\zeta'_k\zeta_{k+}\beta(k+1,\infty)'\} \qquad [A.3]\\
&= A^* - B^*,
\end{aligned}
$$

where $A^\star$ and $B^\star$ denote the first and second terms of the right-hand side of [A.3], respectively. It can be shown that

$$
\|A^*\|^2 = \|m^{-1/2}\{\beta(k+1,\infty)\zeta'_{k+}I\zeta_{k+}\beta(k+1,\infty)'\}\|^2 = O_p(k^{-4d-2}m^{1+4d})
$$

and

$$
\|A^*\| = O_p(k^{-2d-1}m^{0.5+2d}). \qquad [A.4]
$$

To show [A.4], we note that for any conformable matrix A and B,

$$
\|AB\|^2 \le \|A\|^2\ \|B\|^2.
$$

Therefore,

$$
\begin{aligned}
\|A^*\|^2 =& \|m^{-1/2}\{\beta(k+1,\infty)\zeta'_{k+}I\zeta_{k+}\beta(k+1,\infty)'\}\|^2\\
&= m^{-1}\left(\sum_{t=m+1}^{n=bm}\left(\sum_{j=k+1}^{\infty}\beta_j\eta_{t-j}\right)^2\right)^2 \qquad [A.5]
\end{aligned}
$$

We note that

$$
\begin{aligned}
\mathbf{E}\left(\sum_{j=k+1}^{\infty}\beta_j\eta_{t-j}\right)^2 &= \sum_{j=k+1}^{\infty}\beta_j^2\,\mathbf{E}(\eta_{t-j}^2) + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v\,\mathbf{E}(\eta_{t-v}\eta_{t-j})\\
&= \sum_{j=k+1}^{\infty}\beta_j^2\gamma_0 + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v\gamma_{v-j} \qquad [A.6]\\
&\le \sum_{j=k+1}^{\infty}\beta_j^2\gamma_0 + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v|\gamma_{v-j}|.
\end{aligned}
$$

Following the proof of Hosking (1996, p. 277), and letting $v - j = s$ and $x = \frac{s}{m}$, we can show that [A.6] is bounded by

$$
\begin{aligned}
&\sum_{j=k+1}^{\infty} \beta_j^2 \gamma_0 + |C_2| \sum_{j=k+1}^{\infty} \sum_{v=j+1}^{\infty} |\beta_j||\beta_v||(v-j)|^{2d-1} \\
&\leq O_p(k^{-2d-1}) + k^{-d-1} \sum_{j=k+1}^{\infty} \sum_{v=j+1}^{\infty} |(v-j)|^{2d-1} v^{-d-1} \\
&= O_p(k^{-2d-1}) + k^{-d-1} \sum_{j=k+1}^{\infty} \sum_{s=1}^{\infty} s^{2d-1} (s+j)^{-d-1} \\
&\leq O_p(k^{-2d-1}) + k^{-d-1} \sum_{j=k+1}^{\infty} (1+j)^{-d-1} \sum_{s=1}^{\infty} s^{2d-1} \\
&\leq O_p(k^{-2d-1}) + k^{-2d-1} \sum_{s=1}^{\infty} s^{2d-1} \\
&\sim O_p(k^{-2d-1}) + k^{-2d-1} m^{2d} \int_0^1 x^{2d-1} dx \quad \text{as} \quad m \to \infty \\
&= O_p\left(k^{-2d-1} m^{2d}\right),
\end{aligned} \tag{A.7}
$$

where $\beta_j^2 \sim C^2 j^{-2d-2}$, $\gamma_\tau = E(\eta_t \eta_{t+\tau}) \sim C_2 \tau^{2d-1}$ and C and C_2 are finite constants. It follows that

$$\sum_{j=k+1}^{\infty} \beta_j \eta_{t-j} = O_p(k^{-d-0.5} m^d). \tag{A.8}$$

Similarly, we can show that

$$\|B^*\| \leq \|m^{-1/2} \beta(k+1,\infty) \zeta_{k+}'\| \|\zeta_k (\zeta_k' \zeta_k)^{-1} \zeta_k' \zeta_{k+} \beta(k+1,\infty)'\|,$$

where

$$\|m^{-1/2} \beta(k+1,\infty) \zeta_{k+}'\|^2 = m^{-1} \sum_{t=m+1}^{n=bm} \left(\sum_{j=k+1}^{\infty} \beta_j \eta_{t-j} \right)^2 = O_p(k^{-2d-1} m^{2d})$$

and

$$\|m^{-1/2} \beta(k+1,\infty) \zeta_{k+}'\| = O_p(k^{-d-0.5} m^d). \tag{A.9}$$

$$\begin{aligned}
&\|\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\zeta_{k+}\beta(k+1,\infty)'\|^2 \\
&= tr\left(\beta(k+1,\infty)\zeta_{k+}'\zeta_k(\zeta_k'\zeta_k)^{-1}\ \zeta_k'\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\zeta_{k+}\beta(k+1,\infty)'\right) \\
&= \sum_{t=m+1}^{n=bm}\left(\sum_{j=k+1}^{\infty}\beta_j\eta_{t-j}\right)^2 \qquad [A.10] \\
&= O_p(m^{1+2d}k^{-2d-1})
\end{aligned}$$

and

$$\|\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\zeta_{k+}\beta(k+1,\infty)'\| = O_p(m^{1/2+d}k^{-d-0.5}). \qquad [A.11]$$

Thus, by [A.9] and [A.11], it follows that

$$B^* = O_p(m^{2d+0.5}k^{-2d-1}). \qquad [A.12]$$

Furthermore, [A.4] and [A.12] imply that

$$\begin{aligned}
(I) &= m^{-1/2}(\beta(k+1,\infty)\zeta_{k+}'M_k\zeta_{k+}\beta(k+1,\infty)') \\
&= O_p(m^{2d+0.5}k^{-2d-1}). \qquad [A.13]
\end{aligned}$$

The second term (*II*) on the right-hand side of [A.2],

$$\begin{aligned}
(II) &= m^{-1/2}\beta(k+1,\infty)\zeta_{k+}'M_k\varepsilon_{n-m} \\
&= m^{-1/2}\beta(k+1,\infty)\zeta_{k+}'\varepsilon_{n-m} - m^{-1/2}\beta(k+1,\infty)\zeta_{k+}\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m} \\
&= C^* - D^*, \qquad [A.14]
\end{aligned}$$

where C^* and D^* denote the first and second terms of the right-hand side of [A.14], respectively.

Noting that

$$\begin{aligned}
&\|m^{-1/2}\beta(k+1,\infty)\zeta_{k+}'\varepsilon_{n-m}\|^2 \\
&= m^{-1}\|\beta(k+1,\infty)\zeta_{k+}'\|^2\|\varepsilon_{n-m}\|^2 \qquad [A.15] \\
&= O(k^{-2d-1}m^{2d}),
\end{aligned}$$

it follows that

$$C^* = O_p(k^{-d-0.5}m^d). \qquad [A.16]$$

Similarly, by [A.15], we can show that

$$\|D^*\|^2 = \|m^{-1/2}\beta(k+1,\infty)\zeta_{k+}\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m}\|^2$$
$$= m^{-1}tr\left(\varepsilon_{n-m}\zeta_k(\zeta_k'\zeta_k)'^{-1}\zeta_k'\zeta_{k+}'\beta(k+1,\infty)'\beta(k+1,\infty)\zeta_{k+}\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m}\right)$$
$$= O_p(m^{2d}k^{-2d-1}).$$

Thus,

$$(II) = O_p(m^d k^{-d-0.5}) \qquad [A.17]$$

Likewise, the third term (*III*) on the right-hand side of [A.2],

$$(III) = \varepsilon_{n-m}' M_k \zeta_{k+}\beta(k+1,\infty) = O_p(m^d k^{-d-0.5}). \qquad [A.18]$$

To calculate the asymptotic properties of the fifth term (*V*) on the right-hand side of [A.2], we note that by the Theorem 3 of Hosking (1996) and the Theorem 2 of Poskitt (2007), we can show that

$$\|\widehat{\Gamma}_k - \Gamma_k\| = O_p(k(logm/m)^{0.5-d}),\ d \in (0, 0.5), \qquad [A.19]$$

where $\widehat{\Gamma}_k = [\frac{1}{m-k}\sum_{t=k+1}^m \eta_{t+i}\eta_{t+j}]_{i,j=1,2,\cdots,k}$ and $\Gamma_k = [\eta_{t+1}\eta_{t+l}]_{i,j=1,2,\cdots,k}$. This result has also been shown in the proof procedure of Corollary 1 of Poskitt (2007). We further denote that

$$\widehat{\Omega}_k = \frac{\zeta_k'\zeta_k}{n-m}$$
$$= \begin{bmatrix} \frac{1}{n-m}\sum_{i=m+1}^n \eta_{i-1}^2 & \frac{1}{n-m}\sum_{i=m+1}^n \eta_{i-1}\eta_{i-2} & \cdots & \frac{1}{n-m}\sum_{i=m+1}^n \eta_{i-1}\eta_{i-k} \\ \cdots & \cdots & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \frac{1}{n-m}\sum_{i=m+1}^n \eta_{i-k}\eta_{i-1} & \cdots & \cdots & \frac{1}{n-m}\sum_{i=m+1}^n \eta_{i-k}^2 \end{bmatrix}$$

and

$$\Omega_k = \mathbf{E}\left(\zeta_k'\zeta_k\right)$$
$$= \begin{bmatrix} \mathbf{E}\,\eta_{i-1}^2 & \mathbf{E}\,\eta_{i-1}\eta_{i-2} & \cdots & \mathbf{E}\,\eta_{i-1}\eta_{i-k} \\ \cdots & \cdots & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{E}\,\eta_{i-k}\eta_{i-1} & \cdots & \cdots & \mathbf{E}\,\eta_{i-k}^2 \end{bmatrix}$$

Thus,

$$\begin{aligned}\left\|\widehat{\Omega}_k^{-1}\right\| &\leq \left\|\widehat{\Omega}_k^{-1}-\Omega_k^{-1}\right\|+\|\Omega_k^{-1}\|\\ &=\left\|\widehat{\Omega}_k^{-1}(\Omega_k-\widehat{\Omega}_k)\Omega_k^{-1}\right\|+\|\Omega_k^{-1}\|\\ &\leq \left\|\widehat{\Omega}_k^{-1}\right\|\left\|(\Omega_k-\widehat{\Omega}_k)\right\|\|\Omega_k^{-1}\|+\|\Omega_k^{-1}\|\end{aligned}$$

and

$$\|\Omega_k^{-1}\|<\infty,$$

because of a nonsingular matrix Ω_k and by Lemma 5.7 of Poskitt (2007) and Proposition 4.5.3 of Brockwell and Davis (1991). Hence, by [A.19],

$$\left\|\widehat{\Omega}_k^{-1}\right\| \leq \frac{\|\Omega_k^{-1}\|}{(1-\|(\Omega_k-\widehat{\Omega}_k)\|\,\|\Omega_k^{-1}\|)}=O_p(k^{-1}(logm/m)^{d-0.5}). \qquad [A.20]$$

Thus, by [A.19] the fifth term (V) could be

$$\begin{aligned}&\|m^{-1/2}\varepsilon_{n-m}'\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m}\|^2\\ &=\|m^{-1/2}\varepsilon_{n-m}'\left(\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\right)'\varepsilon_{n-m}\|^2\\ &=\|m^{-1}\varepsilon_{n-m}'\left(\zeta_k(\frac{\zeta_k'\zeta_k}{n-m})^{-1}\zeta_k'\right)'\frac{1}{n-m}\varepsilon_{n-m}\|^2\\ &\leq\|m^{-1/2}\varepsilon_{n-m}\zeta_k\|^2\|(\frac{\zeta_k'\zeta_k}{n-m})^{-1}\|^2\|\frac{1}{n-m}\varepsilon_{n-m}\zeta_k'\|^2\\ &=O_p(k)O_p(k^{-2}(logm/m)^{-1+2d})O_p(km^{-1})\\ &=O_p((logm)^{-1+2d}m^{-2d}).\end{aligned} \qquad [A.21]$$

Thus, by [A.21], we note that

$$\begin{aligned}&\frac{1}{m^{1/2}}\varepsilon_{n-m}'M_k\varepsilon_{n-m}\\ &=\frac{1}{m^{1/2}}\varepsilon_{n-m}'\left(I-\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\right)\varepsilon_{n-m}\\ &=\frac{1}{m^{1/2}}\varepsilon_{n-m}'\varepsilon_{n-m}-\frac{1}{m^{1/2}}\varepsilon_{n-m}'\left(\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m}\right)\\ &=\frac{1}{m^{1/2}}\varepsilon_{n-m}'\varepsilon_{n-m}-O_p((logm)^{d-0.5}m^{-d}).\end{aligned} \qquad [A.22]$$

Therefore, when we restrict k on the growth rate of $m^{0.8}$, i.e. $k = O(m^{0.8})$, [A.8], [A.13], [A.17], [A.18], and [A.21] converge to zero. It follows that (I), (II), (III), and (V) converge to zero.

Hence, by [A.13], [A.16], [A.17], [A.18], and [A.22], when $k = O(m^{0.8})$ and $0 < d < 0.5$,

$$\widehat{Q} = m^{-1/2}\varepsilon'_{n-m}\varepsilon_{n-m} + o_p(1). \qquad [A.23]$$

That is, by [A.23], we can obtain

$$m^{-1/2}\widehat{S}_n = 2^{-1/2}m^{-1/2}\sum_{t=m+1}^{n=bm}\left(\frac{\widehat{e}_{t,k}^2}{\widehat{\sigma}_k^2} - \frac{m}{m-2}\right) = m^{-1/2}S_n + o_p(1), \qquad [A.24]$$

when $0 < d < 0.5$ and $k = O(m^{0.8})$.

We now focus on the property of [A.2] for the case of $-0.25 < d < 0$. We first show that

$$\begin{aligned}
\mathbf{E}\left(\sum_{j=k+1}^{\infty}\beta_j\eta_{t-j}\right)^2 &= \sum_{j=k+1}^{\infty}\beta_j^2\,\mathbf{E}(\eta_{t-j}^2) + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v\,\mathbf{E}(\eta_{t-v,i}\eta_{t-j,i}) \\
&= \sum_{j=k+1}^{\infty}\beta_j^2\gamma_0 + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v\gamma_{v-j} \\
&\le \sum_{j=k+1}^{\infty}\beta_j^2\gamma_0 + \sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}\beta_j\beta_v|\gamma_{v-j}|. \\
&\le \sum_{j=k+1}^{\infty}\beta_j^2\gamma_0 + |C_2|\sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}|\beta_j||\beta_v||(v-j)|^{2d-1} \\
&= O_p(k^{-2d-1}) + k^{-d-1}\sum_{j=k+1}^{\infty}\sum_{v=j+1}^{\infty}|(v-j)|^{2d-1}v^{-d-1} \\
&= O_p(k^{-2d-1}) + k^{-d-1}\sum_{j=k+1}^{\infty}\sum_{s=1}^{\infty}s^{2d-1}(s+j)^{-d-1} \\
&\le O_p(k^{-2d-1}) + k^{-d-1}\sum_{j=k+1}^{\infty}(1+j)^{-d-1}\sum_{s=1}^{\infty}s^{2d-1} \\
&\le O_p(k^{-2d-1}) + k^{-2d-1}\sum_{s=1}^{\infty}s^{2d-1} \\
&= O_p\left(k^{-2d-1}\right). \qquad [A.25]
\end{aligned}$$

By the similar manupulation that we used for proving the case of $0 < d < 0.5$, we note that when $-0.25 < d < 0$ and k is set proportional to the sample size m, such as $k = O(m)$, the first, second, and third terms on the right-hand side of [A.2] are

$$m^{-1/2}(\beta(k+1,\infty)\zeta_{k+}' M_k \zeta_{k+}\beta(k+1,\infty)') = O_p(m^{0.5}k^{-2d-1}) \rightarrow 0, \quad [A.26]$$

$$m^{-1/2}\beta(k+1,\infty)\zeta_{k+}' M_k \varepsilon_{n-m} = O_p(k^{-2d-1}) \rightarrow 0, \quad [A.27]$$

$$\varepsilon_{n-m}' M_k \zeta_{k+}\beta(k+1,\infty) = O_p(m^d k^{-d-0.5}) \rightarrow 0. \quad [A.28]$$

Likewise, to prove the asymptotic properties of the fifth term (V) on the right-hand side of [A.2], by the Theorem 3 of Hodking (1996) and the Theorem 2 of Poskitt (2007), we note that [A.19] becomes

$$\|\widehat{\Gamma}_k - \Gamma_k\| = O_p(k(logm/m)^{0.5}), \quad d \in (-0.5, 0). \quad [A.29]$$

Thus,

$$\begin{aligned}
&\|m^{-1/2}\varepsilon_{n-m}'\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\varepsilon_{n-m}\|^2 \\
&= \|m^{-1/2}\varepsilon_{n-m}'\left(\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\right)'\varepsilon_{n-m}\|^2 \\
&= \|m^{-1/2}\varepsilon_{n-m}'\left(\zeta_k(\frac{\zeta_k'\zeta_k}{n-m})^{-1}\zeta_k'\right)'\frac{1}{n-m}\varepsilon_{n-m}\|^2 \\
&\le \|m^{-1/2}\varepsilon_{n-m}\zeta_k\|^2\|(\frac{\zeta_k'\zeta_k}{n-m})^{-1}\|^2\|\frac{1}{n-m}\varepsilon_{n-m}\zeta_k'\|^2 \\
&= O_p(k)O_p(k^{-2}(logm/m)^{-1})O_p(km^{-1}) \\
&= O_p((logm)^{-1}).
\end{aligned} \quad [A.30]$$

and

$$\begin{aligned}
&\frac{1}{m^{1/2}}\varepsilon_{n-m}'M_k\varepsilon_{n-m} \\
&= \frac{1}{m}^{-1/2}\varepsilon_{n-m}'\left(I-\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k'\right)\varepsilon_{n-m} \\
&= \frac{1}{m^{1/2}}\varepsilon_{n-m}'\varepsilon_{n-m} - \frac{1}{m^{1/2}}\varepsilon_{n-m}'\left(\zeta_k(\zeta_k'\zeta_k)^{-1}\zeta_k')\varepsilon_{n-m}\right) \\
&= \frac{1}{m^{1/2}}\varepsilon_{n-m}'\varepsilon_{n-m} - O_p((logm)^{-0.5}).
\end{aligned} \quad [A.31]$$

Then, by [A.26], [A.27], [A.28], and [A.31], when $k = O(m)$ and $-0.25 < d < 0$,

$$\widehat{Q} = m^{-1/2}\varepsilon_{n-m}'\varepsilon_{n-m} + o_p(1). \quad [A.32]$$

That is, by [A.32], we can obtain

$$m^{-1/2}\widehat{S}_n = 2^{-1/2}m^{-1/2}\sum_{t=m+1}^{n=bm}\left(\frac{\widehat{e}_{t,k}^2}{\widehat{\sigma}_k^2} - \frac{m}{m-2}\right) = m^{-1/2}S_n + o_p(1), \qquad [A.33]$$

when $-0.25 < d < 0$ and $k = O(m)$.

As a consequence, by [A.24] and [A.33], we could conclude that when $d \in (-0.25, 0) \cup (0, 0.5)$ and $k = O(m)$,

$$m^{-1/2}\widehat{S}_n = 2^{-1/2}m^{-1/2}\sum_{t=m+1}^{n=bm}\left(\frac{\widehat{e}_{t,k}^2}{\widehat{\sigma}_k^2} - \frac{m}{m-2}\right) = m^{-1/2}S_n + o_p(1), \qquad [A.34]$$

A.2 Proof of Theorem 1

By the results of our Lemma 2, the Theorem 1 of Inclan and Tiao (1994), the Theorem 1 can be established immediately.

A.3 Proof of Theorem 2

We first do the first difference on y_t, i.e. $\Delta y_t = \eta_t$. Hence, η_t can be approximated by an AR(k) model as

$$\eta_t = \sum_{i=1}^{k}\widehat{b}_j\eta_{t-j} + \widehat{e}_{tk},$$

where the growth rate of k satisfies the Lemma 1. Then, by our Lemma 2, the Theorem 1 of Inclan and Tiao (1994) and following the same reasoning of Theorem 1, we can build up the Therorem 2.

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